



Vectors: Diagrams and Proof 1i

Subject & topics: Maths Stage & difficulty: A Level P2

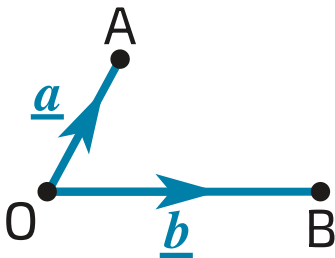


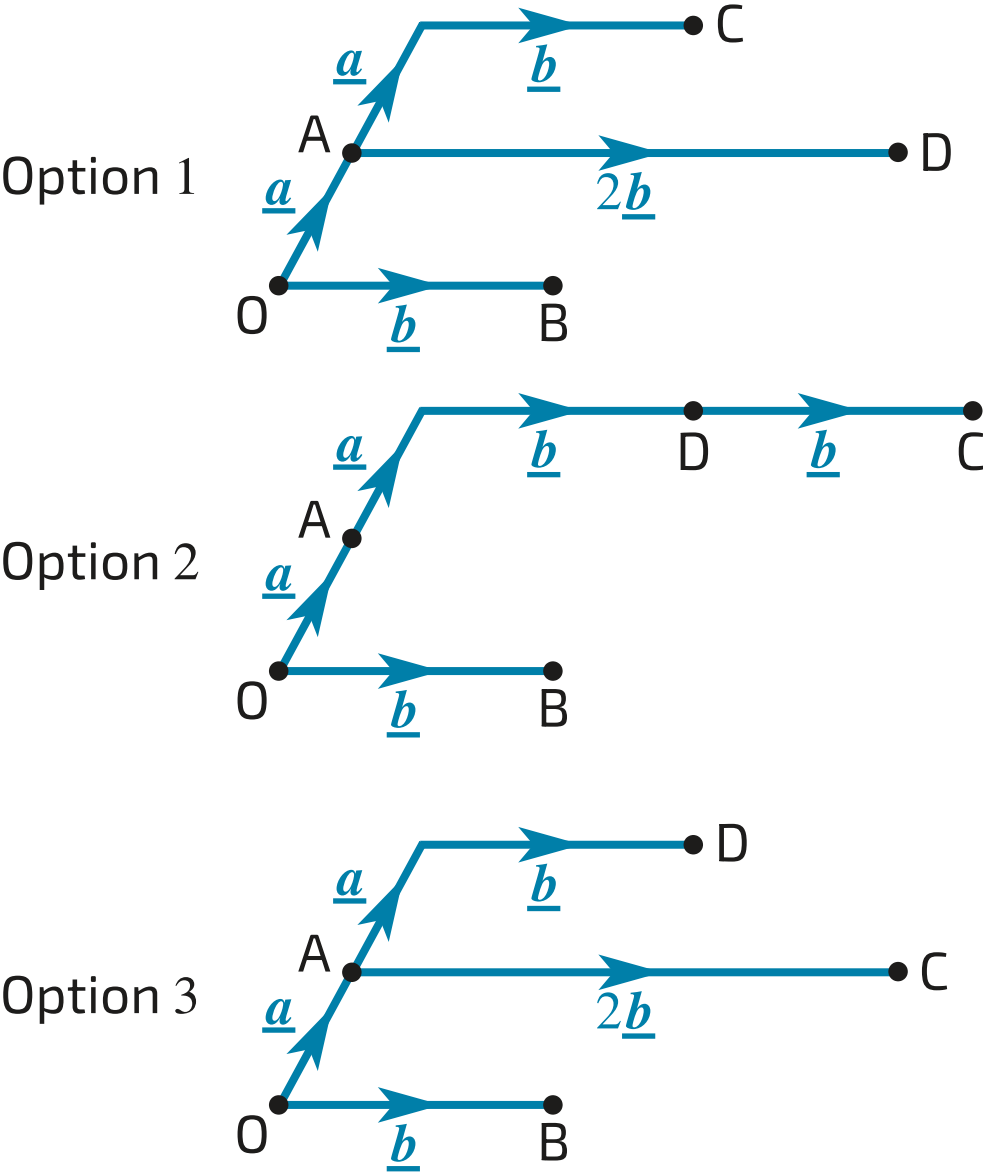
Figure 1: Three points O, A and B, and vectors joining them.

Figure 1 shows points O, A and B, with $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OB} = \underline{b}$.

Part A
Sketch

Make a sketch of the diagram and mark the points C and D such that $\overrightarrow{OC} = \underline{a} + 2\underline{b}$ and $\overrightarrow{OD} = 2\underline{a} + \underline{b}$.

Choose the correct sketch from the 3 options below.



- ☐ Option 1
- ☐ Option 2
- ☐ Option 3

Part B

Vector $\overrightarrow{DC}$

Express $\overrightarrow{DC}$ in terms of $\underline{a}$ and $\underline{b}$, simplifying your answer.

$\overrightarrow{DC} =$ $\underline{a} +$ $\underline{b}$

Part C

Proof

Prove that ABCD is a parallelogram.

Sides $\overrightarrow{AD}$ and are both equal to . Therefore they are parallel and of equal length.

Sides $\overrightarrow{AB}$ and are both equal to . Therefore they are parallel and of equal length.

The quadrilateral ABCD has two pairs of parallel sides of equal length. Therefore ABCD is a parallelogram.

Items:

$\underline{b} - \underline{a}$

$\overrightarrow{BC}$

$\overrightarrow{CD}$

$\overrightarrow{DC}$

$\overrightarrow{CB}$

$\underline{a} - \underline{b}$

$\underline{a} + \underline{b}$

$-\underline{a} - \underline{b}$

Adapted with permission from UCLES, A Level, Jan 2003

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3D Vectors 2i

Subject & topics: Maths **Stage & difficulty:** A Level P2

ABCD is a quadrilateral. You are given four pieces of information:

Relative to a fixed origin O , the position vector of A is $2\underline{i} + 5\underline{j} + 8\underline{k}$.

Relative to a fixed origin O , the position vector of B is $5\underline{i} + 9\underline{j} + 8\underline{k}$.

The vector $\overrightarrow{BC} = \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix}$.

The vector $\overrightarrow{BD} = \begin{pmatrix} -3 \\ -4 \\ 5 \end{pmatrix}$.

Part A

Finding $\overrightarrow{AB}$

Find the vector $\overrightarrow{AB}$. Give your answer in $\underline{i}, \underline{j}, \underline{k}$ form.

$$\overrightarrow{AB} = \text{ } \underline{i} + \text{ } \underline{j} + \text{ } \underline{k}$$

Part B

Finding $\overrightarrow{CD}$

Find the vector $\overrightarrow{CD}$. Give your answer in $\underline{i}, \underline{j}, \underline{k}$ form.

$$\overrightarrow{CD} = \text{ } \underline{i} + \text{ } \underline{j} + \text{ } \underline{k}$$

Part C

Finding $\overrightarrow{AD}$

Find the vector $\overrightarrow{AD}$. Give your answer in $\underline{i}, \underline{j}, \underline{k}$ form.

$\overrightarrow{AD} =$ $\underline{i} +$ $\underline{j} +$ $\underline{k}$

Part D

Type of quadrilateral

The shape ABCD lies in a plane. What type of quadrilateral is ABCD?

- ☐ An irregular quadrilateral
- ☐ A trapezium
- ☐ A rhombus
- ☐ A kite
- ☐ A rectangle
- ☐ A parallelogram
- ☐ A square

Created for isaacphysics.org by Jonathan Waugh

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Constant Acceleration 2i

Subject & topics: Maths **Stage & difficulty:** A Level P1

A stone is released from rest on a bridge and falls vertically into a lake. The stone has velocity 14 m s^{-1} when it enters the lake.

Part A

Distance fallen

Calculate the distance the stone falls before it enters the lake.

Part B

Time on entering the lake

Calculate the time after its release when it enters the lake. Give your answer to 3 significant figures.

Part C

Acceleration

The lake is 15 m deep and the stone has velocity 20 m s^{-1} immediately before it reaches the bed of the lake.

Given that there is no sudden change in the velocity of the stone when it enters the lake, find the acceleration of the stone while it is falling through the lake. Give your answer to 2 significant figures.

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Kinematics Graphs

Subject & topics: Maths **Stage & difficulty:** A Level C2

A man drives a car on a horizontal straight road. At $t = 0$ where the time t is in seconds, the car runs out of petrol. At this instant the car is running at 12 m s^{-1} . The car decelerates uniformly, coming to rest when $t = 8$. The man then walks back along the road at 0.7 m s^{-1} until he reaches a petrol station a distance of 420 m from his car. After his arrival at the petrol station it takes him 250 s to obtain a can of petrol. He is then given a lift back to his car on a motorcycle. The motorcycle starts from rest and accelerates uniformly until its speed is 20 m s^{-1} ; it then decelerates uniformly, coming to rest at the stationary car at time $t = T$.

Part A

(t, v) graph

Sketch the shape of the (t, v) graph for the man for $0 \leq t \leq T$. [Your sketch need not be drawn to scale yet numerical values should be shown].

To see an example sketch, answer the following question: What is the largest positive value of v on the graph?

Part B

Finding deceleration

Find the deceleration of the car for $0 < t < 8$.

Part C
Finding T

Find the value of T .

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Projectiles: General 3ii

Subject & topics: Maths **Stage & difficulty:** A Level P2

A golfer hits a ball from a point O on horizontal ground with a velocity of 55 m s^{-1} at an angle of 20° above the horizontal. The ball first hits the ground at a point A and the time of flight is t seconds. Assume that there is no air resistance.

Part A

Value of t

Calculate the value of t to 3 significant figures.

Part B

Distance OA

Calculate the distance OA to 3 significant figures.

Part C

Speed of motion

Calculate the speed of motion of the ball 2.6 s after the golfer hits the ball. Give your answer to 3 significant figures.

Part D

Direction of motion

Calculate the direction of motion of the ball 2.60 s after the golfer hits the ball. Give your answer, measured below the horizontal, to 3 significant figures.

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Applying Trigonometry

Subject & topics: Maths **Stage & difficulty:** A Level C2

A rower attempts to row across a river from a place **A** where the banks of the river are straight and parallel. They wish to reach a point **B** which is directly opposite to **A** on the other bank (i.e. on a line perpendicular to the bank at point **A**).

They start to row towards **B**, keeping the boat aligned in a direction parallel to $\underline{j}$, but discover that there is a current flowing in a direction $\underline{i}$ parallel to the banks, such that their resultant travel is along a vector $\underline{v} = \underline{i} + 4\underline{j}$.

Part A

Speed of the rower

Find the magnitude of vector $\underline{v}$.

The following symbols may be useful: v



Part B

Angle between $\underline{v}$ and $\underline{i}$

Find the angle between vectors $\underline{v}$ and $\underline{i}$. Give your answer to no more than 3 sf.

Part C

Direction of travel

If they are to arrive at B, but can adjust their rowing speed to cross to the other bank in the same time that it would take if there were no current, in what direction should they actually row? Give your answer as a vector in terms of the unit vectors i and j .



The following symbols may be useful: i , j , v_{row}

What is the angle between $\underline{v}$ and $\underline{j}$? Give your answer to no more than 3 sf.

Part D

A tower in the distance

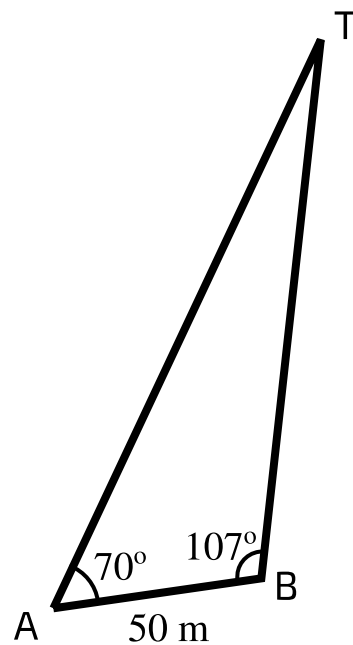


Figure 1: The tower, and points A and B.

Some walkers see a tower, T, in the distance and want to know how far away it is. They take a bearing from point A and walk for 50 m in a straight line before taking another bearing from point B. They find that the angle TAB is 70° and angle TBA is 107° (see **Figure 1**).

Find the distance of the tower from A. Give your answer to 3 sf.

Part E**Distance from C**

They continue walking in the same direction for another 100 m to a point C, so that AC is 150 m. What is the distance of the tower from C? Give your answer to 3 sf.

Part F**Shortest distance from A to C**

Find the shortest distance of the walkers from the tower as they walk from A to C. Give your answer to 3 sf.

Part G**Area swept out**

D is the point on AC such that TD is the shortest distance of the walkers from the tower.

Find the area of ground represented by the triangle ATD. Give your answer in km^2 and to 3 sf.

Adapted with permission from SAW 2017 and UCLES, A Level, January 2009, Paper 4722, Question 5.

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Trigonometry: Double Angles 4i

Subject & topics: Maths

Stage & difficulty: A Level P2

Part A

Proof

By first expanding $\cos(2\theta + \theta)$, rewrite $\cos 3\theta$ in terms of $\cos \theta$.

The following symbols may be useful: $\cos()$, θ

Part B

$\cos 6\theta$

Hence write $\cos 6\theta$ in terms of $\cos \theta$.

The following symbols may be useful: $\cos()$, θ

Part C

The equation $1 + \cos 6\theta = 18 \cos^2 \theta$

Give the smallest positive solution to the equation

$$1 + \cos 6\theta = 18 \cos^2 \theta$$

in degrees.

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Trigonometry: Combined Angles 4ii

Subject & topics: Maths Stage & difficulty: A Level C1

Part A

Combined Angles

Express $8 \sin \theta - 6 \cos \theta$ in the form $R \sin (\theta - \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$.

Give the exact value of R .

$R =$

Give the value of α to three significant figures.

$\alpha =$

Part B

Solve

Hence solve, for $0^\circ < \theta < 360^\circ$, the equation $8 \sin \theta - 6 \cos \theta = 9$, giving your answers in degrees to three significant figures.

()

Part C

Maximum Value

Hence find the greatest possible value of

$$32 \sin x - 24 \cos x - (16 \sin y - 12 \cos y)$$

as the angles x and y vary.

If the value is not an integer, enter the value as a decimal, giving your answer to 3 sf if necessary.

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